

Lab03: Real-World Data Linear Regression

DATA

The data represents below approved building permit statistics categorized by region or district, detailing construction metrics such as total number of buildings, total floor area, and total construction value. It captures regional/district observations mapping the Total Floor Area (m²) against the Total Construction Value (PHP '000).

- Independent Variable (x): Total Floor Area (m²)
- Dependent Variable (y): Total Construction Value (PHP '000)

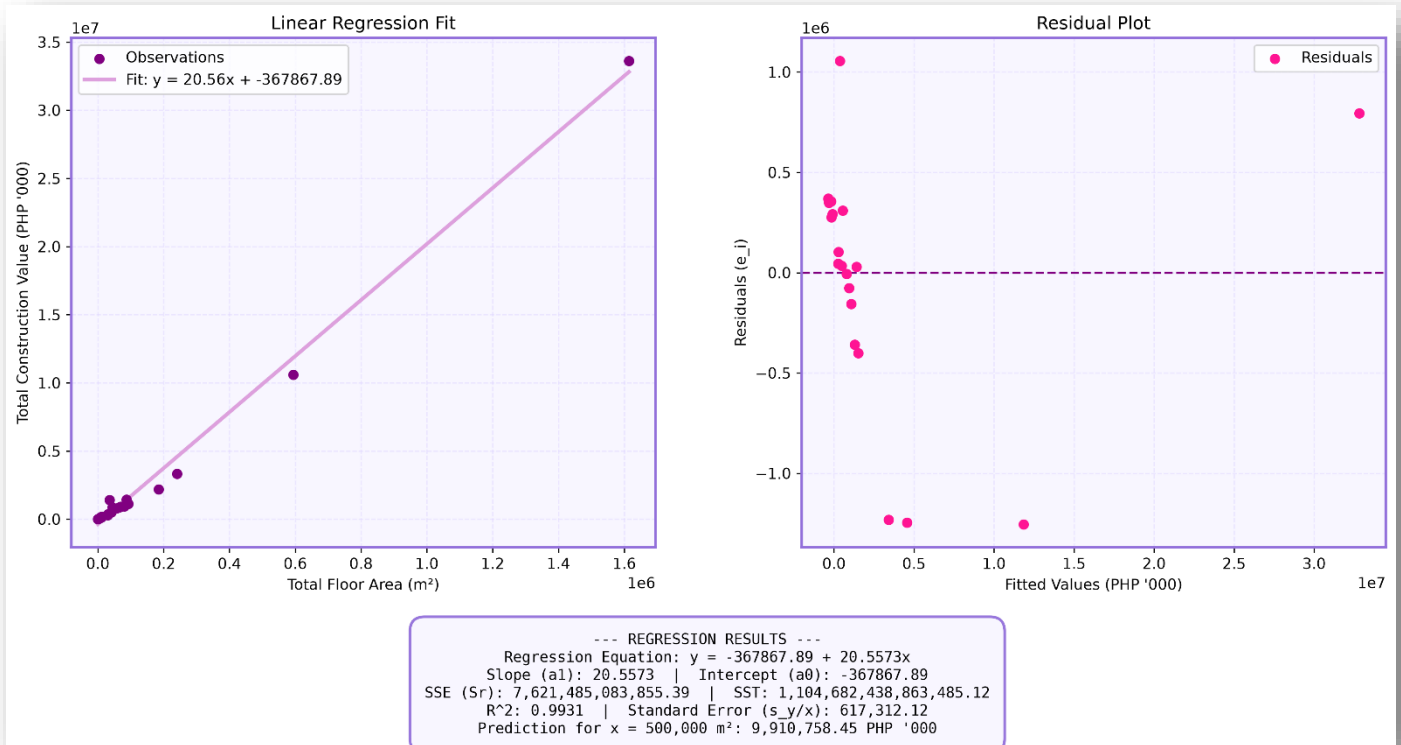
Table of Observations

No.	Location / District (x label)	Total Floor Area (x, m ²)	Total Construction Value (y, PHP '000)
1	Tondo	184,402	2,190,872
2	Binondo	92,176	1,125,793
3	Quiapo	40,652	502,476
4	San Nicolas	0	947
5	Sta. Cruz	31,459	381,895
6	Sampaloc	56,334	784,587
7	San Miguel	30,366	301,482
8	Ermita	64,117	873,698
9	Intramuros	3,777	61,073
10	Malate	9,151	175,137
11	Paco	9,601	104,693
12	Pandacan	2,681	35,594
13	Port Area	350	7,897
14	Sta. Ana	12,799	187,871
15	City of Mandaluyong	35,526	1,417,072
16	City of Marikina	86,667	1,442,899

17	City of Pasig	594,289	10,595,385
18	Quezon City	1,614,127	33,608,497
19	City of San Juan	44,529	857,648
20	City of Caloocan	240,383	3,327,288
21	City of Malabon	70,325	921,442
22	City of Navotas	80,660	931,450

- The dataset was obtained from the Philippine Statistics Authority (PSA) OpenSTAT Database
- Full link:
https://openstat.psa.gov.ph/PXWeb/pxweb/en/DB/DB__2G__CONS__BGP/?tablelist=true

REGRESSION RESULT



Interpret the Results

What the slope and the intercept mean:

- **Slope (20.56):** For every extra 1 m² of floor space, the construction cost goes up by about 20.56 thousand pesos (PHP 20,557.30).
- **Intercept (-367,867.89):** This is just the starting point mathematically to make the line fit the graph properly. Even though a negative cost doesn't happen in real life, it helps the formula line up correctly with the data.

How well the line fits (R^2 and standard error):

- R^2 (**0.9931**): This shows that floor size and construction cost match up almost perfectly. About 99.31% of the cost is explained by how big the floor area is.
- **Standard Error (617,312.12):** This tells us the average amount that real-world costs might stray away from our prediction line.

What the residuals show:

- Residuals are the gaps between the actual data points and our trend line. Because they are spread out evenly, it proves that using a straight trend line is a good choice for this data.

Prediction

Let's predict the construction cost for a building project with a floor area of **500,000 m²**

Substitution and Result:

$$\text{Substitution: } y = -367,867.89 + (20.5573 \times 500,000)$$

$$\text{Result: } y = 9,910,758.45 \text{ PHP '000}$$

Why it is meaningful:

- This final result is roughly PHP 9.91 billion. It is useful because it gives planners and government agencies a quick, reliable way to estimate how much a large construction project will cost based only on its floor size.